SWA-PTIS: Socially-Weighted Alignment with Prospect-Theory Importance Sampling for Cross-Cultural Moral Alignment of Large Language Models

Anonymous Authors
anonymous@institution.edu

Abstract

Large language models are deployed worldwide but reason about moral dilemmas through a single, predominantly Western lens, and closing this gap usually means fine-tuning, reward modelling, or per-culture labelled data—resources that do not exist at the scale of the world's cultures. We ask whether a frozen model can instead be nudged toward country-specific moral preferences at inference time, using only what cross-cultural surveys already tell us. We propose **SWA-PTIS** (Socially-Weighted Alignment with Prospect-Theory Importance Sampling), which equips the model with a small panel of culturally-grounded personas derived from World Values Survey data and applies a single KL-regularised importance-sampling correction in logit space whose utility is evaluated by a cooperative Prospect-Theory value function. The update self-attenuates when personas agree—logically producing no correction in the limit of zero disagreement—without requiring an explicit gating rule. We evaluate on MultiTP across 15 countries, 12 native languages, and six moral dimensions, using five open-weight models from 7B to 72B. On the two strong instruction-tuned 70B-class backbones in our panel (Qwen2.5-72B, Llama-3.1-70B), mean Jensen–Shannon distance to country-level human preferences drops by 0.032 in absolute terms on each model (-37.6% on Qwen2.5-72B, -34.7% on Llama-3.1-70B), exceeding every prompting and activation-steering baseline we tried, while smaller 4-bit-quantised models show consistent directional gains. Mechanistic ablations against deterministic shifts and a binary ARGS variant point to one load-bearing ingredient: the Prospect-Theory gain/loss asymmetry operating on KL-regularised perturbations. Code: `[repository]`.

1 Introduction

When a self-driving car must choose between sparing a group of elderly men or a group of young women, there is no culturally universal answer. The Moral Machine experiment [Awad et al., 2018]-40 million judgments from 233 countries-revealed that such preferences vary profoundly across cultures: East Asian respondents weight youth more heavily; Southern-hemisphere populations show distinct utilitarian calculi; Islamic societies prioritise different social roles. Yet the large language models increasingly enlisted to assist with such decisions encode a single, culturally monolithic moral stance inherited from their predominantly Western training data [Henrich et al., 2010].

The multilingual successor benchmark MultiTP [Jin et al., 2025] operationalises this evidence into country-specific Average Marginal Component Effects (AMCE) across six moral dimensions and 107 languages, providing a rigorous test bed for cross-cultural alignment. The verdict is stark: Jin

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

et al. [2025] found that none of 19 contemporary LLMs achieved broadly satisfactory alignment-most exhibit WEIRD-biased reasoning that departs markedly from non-WEIRD human preferences.

How should this gap be closed? Existing approaches each sacrifice critical properties. *Fine-tuning* [Bai et al., 2022, Ouyang et al., 2022] demands per-culture labelled data and modifies weights irreversibly. *Prompting-based steering* relies on ad-hoc instructions with no empirical grounding [Jin et al., 2025]. *Activation steering* [Arditi et al., 2025, Zou et al., 2023] requires white-box access and per-model engineering. *Reward-guided decoding* [Khanov et al., 2024, Mudgal et al., 2024] needs trained reward models unavailable for cross-cultural settings. *Culture-aware adapters* [Li et al., 2024, Yao et al., 2024] scale poorly across 100+ countries. None of these methods can adapt an LLM's moral reasoning to a target culture *at inference time* without labelled data or weight modification.

We propose **SWA-PTIS** (Socially-Weighted Alignment with Prospect-Theory Importance Sampling), built on a simple insight: *disagreement is signal*. Moral preferences within any society are not monolithic-age cohorts and social strata exhibit systematically different priorities [Awad et al., 2018, Kleiman-Weiner et al., 2017]. The World Values Survey [Haerpfer et al., 2022] provides empirically calibrated data on these within-country differences across 80+ nations. When culturally grounded persona agents derived from WVS data *disagree* on a moral decision, that disagreement reveals the decision is culturally contested-and can be exploited through stochastic optimal control [Williams et al., 2018] to produce a more culturally representative output.

Concretely, SWA-PTIS constructs $N = 4$ WVS-grounded personas per country (three age cohorts plus a utilitarian agent), evaluates all $N + 1$ agents in a single batched forward pass with native-language prompting across 12 languages, and extracts decision logits. If inter-agent variance exceeds a calibrated threshold τ, it samples $K = 128$ perturbations and scores each via a Prospect Theory value function [Kahneman and Tversky, 1979] that captures the asymmetric gain-loss sensitivity characteristic of human moral judgment. The importance-weighted MPPI update yields a minimal-KL correction; a positional debiasing step cancels recency effects [Liu et al., 2024].

Contributions.

- **A training-free framework** (§3) that turns within-country WVS variation into a usable steering signal: persona disagreement gates a Prospect-Theory-weighted MPPI correction in logit space, applied only when the decision is culturally contested.
- **Empirical evidence on six models** (7B–126B; §5) showing that the largest, cleanest gains come on the well-calibrated 70B-class backbones, while smaller and stronger models exhibit distinct regimes that we report and analyse rather than aggregate.
- **A mechanism analysis backed by a robustness suite** (§6, Appendices G–Q). Ablations isolate the utilitarian anchor and Prospect-Theory asymmetry as the load-bearing components. Deterministic consensus-shift baselines and a Binary-ARGS adaptation then show that WVS weighting alone is insufficient and that the Prospect-Theory gain/loss kernel is the active ingredient. The remaining sweeps-held-out τ calibration, three temperature sweeps, and six dataset preprocessing configurations-all stay within the bootstrap noise floor, ruling out the obvious confounders.

2 Related Work

Cross-cultural moral benchmarks. The Moral Machine experiment [Awad et al., 2018] provides 40M+ cross-cultural moral judgments. MultiTP [Jin et al., 2025] extends this to 107 languages with country-specific AMCEs-we adopt it as our benchmark and propose an alignment *intervention* rather than diagnostics alone. Other benchmarks (ETHICS [Hendrycks et al., 2021], MoralExceptQA [Jin et al., 2022], MoCa [Nie et al., 2023]) remain English-centric. PRISM [Kirk et al., 2024] and GlobalOpinionQA [Durmus et al., 2023] address diversity but cover few languages. Recent frameworks (EvalMORAAL [Aakanksha et al., 2024], CARB [Myung et al., 2024]) emphasise broader-topic evaluations; SWA-PTIS targets binary moral decisions specifically.

Cross-cultural LLM evaluation. LLM outputs correlate most strongly with liberal, educated American respondents [Santurkar et al., 2023]. RLHF introduces unintended cultural impacts [Ryan et al., 2024], and assigning cultural personas can surface toxic outputs [Deshpande et al., 2023]. This recurring WEIRD bias motivates SWA-PTIS's calibration objective.

Inference-time alignment. Activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access. ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] need trained reward models-unavailable for cross-cultural settings (Appendix I). CultureManager [Li et al., 2024] and CAReDiO [Yao et al., 2024] train culture-specific adapters but require per-culture labelled data and weight modification, scaling poorly across 100+ countries. To the best of our knowledge, SWA-PTIS is the first training-free method targeting country-level moral calibration specifically. Sorensen et al. [2024] articulate the goal of pluralistic alignment, which SWA-PTIS operationalises concretely.

Positioning against adjacent inference-time methods. Recent work on inference-time alignment falls into several families that we deliberately do not compete with head-to-head. *DIFFPO*-style policy-agnostic aligners require an additional learned aligner model on top of the base LLM, trading deployment cost for flexibility; *SWAI*-style training-free logit steering targets attribute-level shifts (style, toxicity) via fixed steering vectors, not country-conditioned moral preferences; *SLED*-style self-logit evolution is a white-box factuality method that operates on internal layers. SWA-PTIS is positioned narrowly within this design space: black-box, single-model, binary forced-choice moral decisions with country-conditioned persona grounding, and no extra trained aligner. Relative to structured multi-agent pipelines such as OG-MAR [Zhang et al., 2024], SWA-PTIS trades ontology-level reasoning depth for the much lower deployment cost of one additional batched forward pass.

MPPI and Prospect Theory. MPPI [Williams et al., 2018] is a sampling-based stochastic optimal control algorithm originally developed for robotics; Huang et al. [2024] adapted it for token-level LLM steering. We use MPPI in logit space and combine it with a Prospect-Theory utility [Kahneman and Tversky, 1979] to capture the asymmetric gain/loss sensitivity characteristic of human moral judgment.

3 Method: SWA-PTIS

3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `A` (a) and `B` (b). For target country c, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE vector from 40M+ judgments [Awad et al., 2018]. A frozen LLM f_θ produces logits $\mathbf{z}(\mathbf{x}) = [z_a,\, z_b]$ at the decision token position; the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ where $\Delta = z_b - z_a$. Each dimension's alignment is measured via a **mean preference rate** (MPR):

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

following the MultiTP evaluation convention [Jin et al., 2025]. For the five *binary* dimensions (Species, Gender, Age, Fitness, Social Value), each scenario's grouping indicator takes values $\{0, 1\}$; under the balanced conjoint design [Awad et al., 2018], the empirical mean $\hat{m}_d^{(c)}$ equals the sample analogue of the classical Average Marginal Component Effect (AMCE), the probability that the "preferred" group is spared [Awad et al., 2018]. For **Utilitarianism**, the treatment variable n_{diff} (the numerical life-count difference) is a multi-valued ordered attribute, not a binary indicator. Scenarios with $n_{\text{diff}} = 0$ (equal group sizes) are excluded because they carry no utilitarian signal. The proper conjoint AMCE for a continuous treatment is the regression slope $\hat{b}$ from fitting $p_{\text{spare}} = a + b\, n_{\text{diff}}$ (the marginal change per additional life). We instead follow the MultiTP LLM-evaluation convention and report the *mean preference rate* $\hat{m}_{\text{Util}}^{(c)} = \text{mean}(p_{\text{spare}})$ over remaining scenarios (Appendix A). This is a valid proxy for utilitarian tendency—under the near-balanced distribution of n_{diff} values in Moral Machine, $\hat{m}_{\text{Util}}$ is a monotone function of $\hat{b}$ (Appendix A)—but it is not a causal marginal effect. We use it because (i) it is bounded in $[0, 1]$, (ii) it is the quantity used to construct the MultiTP human ground-truth to which we compare, and (iii) the Pearson-r and JSD alignment metrics are defined over the full six-element MPR vector, making internal consistency the priority.

Goal: Without modifying θ, find a logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$ under Jensen–Shannon Divergence.

3.2 Cultural Persona Construction

For each country c, SWA-PTIS constructs $N = 4$ persona agents from WVS Wave 7 individual-level microdata [Haerpfer et al., 2022]. We extract 8 value dimensions from specific WVS items: *gender egalitarianism* (Q58P, Q59P, Q60P), *religious importance* (Q6P), *interpersonal trust* (Q43P), *moral permissiveness* (Q50, Q52P, Q54P), *work centrality* (Q5P), *family importance* (Q1P), *personal autonomy* (Q39P), and *meritocratic orientation* (Q40P). For each country, respondents are segmented into three age cohorts using birth year and survey year: *young* (18–35), *middle-aged* (36–55), and *older* (>55). Dimension means are computed per cohort and converted into natural-language descriptors via fixed thresholds (Appendix B). A fourth, country-invariant *utilitarian agent* maximises lives saved. Each persona is a natural-language system prompt:

> *You are a [age-cohort] adult from [country]. The cultural values typical of your social group are: [WVS statistics]. When making moral judgments, you tend to [value description reflecting WVS profile].*

WVS grounding ensures empirically measured within-country variation rather than stereotypes [Deshpande et al., 2023]. Concrete persona examples and the WVS-to-trolley dimension linkage are detailed in Appendix B.

3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures T_{cat} (Table 1) scale logits before computing decision gaps:

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N}\sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

Here δ_{base} is the decision gap of the base prompt (no persona). The reward r_i measures how strongly persona i shifts the model away from the *base model output*: a large $|r_i|$ indicates that the persona has identified a direction of moral preference the vanilla model does not exhibit. We note that δ_{base} is not a culturally neutral anchor—RLHF-tuned base models encode WEIRD-skewed priors [Santurkar et al., 2023]—so r_i measures persona deviation from a biased baseline rather than from a neutral one. This is intentional: SWA-PTIS's goal is to *correct* the base model, so personas that agree with it ($|r_i| \approx 0$) provide no corrective signal and are appropriately downweighted.

3.4 Persona Disagreement Monitoring

In the current implementation, MPPI runs unconditionally on every scenario. The per-agent persona variance

$$\widehat{\text{Var}}(\delta_1, \ldots, \delta_N) \tag{5}$$

is computed and recorded as a diagnostic: it quantifies how much the personas disagree about the preferred direction for each scenario. A high variance indicates strong persona conflict (the MPPI correction draws on a wider spread of importance weights), while a low variance indicates persona consensus (the MPPI correction is small, leaving the result near $\bar{\delta}$). Adaptive gating—running MPPI only when variance exceeds a calibrated threshold $\tau^{(c)}$—is a natural extension. A threshold calibrated at the 65th percentile of variances on $n_{\text{cal}} = 50$ held-out scenarios would target a 35% intervention rate. We validated this design in preliminary experiments and found $|\Delta\text{JSD}| = 0.0006$ between the gated and ungated variants across 15 countries (Appendix G), confirming that unconditional MPPI is a safe default: the importance-weighted update self-attenuates when personas agree (the optimal δ^* collapses toward zero), making an explicit gate redundant in practice. We note that with $N = 4$ personas, per-scenario variance estimates are inherently noisy: small-sample estimates with four observations have high relative uncertainty regardless of the underlying distribution. This noise is compounded by cross-persona correlation—all four δ_i's are produced by the same frozen base model on the same prompt, differing only in the persona prefix, so they share substantial structural dependence. The self-attenuating property of the IS update is therefore doubly beneficial: it avoids hard reliance on unreliable per-scenario variance for gating, and the ungated variant empirically matches the gated one to within $|\Delta\text{JSD}| = 0.0006$ (Appendix G).

3.5 Prospect-Theory Importance Sampling (KL-IS)

Relationship to MPPI. We use a 1-step, 1-dimensional instantiation—not full MPPI. MPPI [Williams et al., 2018] is a multi-horizon stochastic optimal-control algorithm that samples K trajectories $\{\epsilon_{k,1}, \ldots, \epsilon_{k,T}\}$ over T time steps and computes path-integral importance weights over the full horizon. Our setting has a scalar state δ and a one-step horizon ($T = 1$). In this degenerate case the trajectory collapses to a single perturbation ϵ_k, the path integral reduces to a standard importance-sampling estimator, and the free-energy KL regulariser reduces to the per-sample log importance ratio (Appendix C). The resulting algorithm is therefore exactly *KL-regularised importance sampling* (KL-IS): draw K samples from a Gaussian prior, score each with a utility, and return the softmax-weighted sum. This is an established information-theoretic control primitive; the "SWA" prefix captures the cooperative social aggregation over N persona agents, and "PTIS" reflects the Prospect-Theory utility and the IS mechanism. We retain citations to Williams et al. [2018] specifically because the derivation of the KL penalty and the importance-weight formula are *identical* to theirs; only the dimensionality (1D, 1-step) differs. Appendix C gives the full algebraic reduction.

SWA-PTIS draws $K = 128$ perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma_k^2)$ and evaluates a collective welfare score for each $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$. The noise standard deviation σ_k is set *adaptively* per scenario: $\sigma_k = \mathrm{std}(\delta_1, \ldots, \delta_N)$ (the inter-agent logit spread), ensuring the perturbation magnitude is commensurate with actual persona disagreement across all models and categories. A fixed fallback $\sigma = 0.3$ is used when all agents agree exactly ($\sigma_k < 10^{-6}$), in which case the IS update self-attenuates regardless.

Per-agent cooperative utility. Define the *normalised agreement score* for persona i at perturbation k as

$$a_{i,k} \;=\; \frac{r_i\,(\tilde{\delta}_k - \delta_{\text{base}})}{\sigma_k^2}, \tag{6}$$

where $\sigma_k = \mathrm{std}(\delta_1, \ldots, \delta_N)$ is the adaptive noise standard deviation. Dividing by σ_k^2 makes $a_{i,k}$ *dimensionless*: both $r_i = \delta_i - \delta_{\text{base}}$ and $\tilde{\delta}_k - \delta_{\text{base}} = \bar{r} + \epsilon_k$ are logit-gap quantities on the scale of σ_k, so their product has units of σ_k^2; the normalisation yields a scale-free signal-to-noise ratio that is comparable across model families and logit-temperature settings. The score is positive when the perturbation moves in the same direction persona i wishes to move, and negative otherwise. The per-agent cooperative utility is then

$$U_i(\epsilon_k) = (1 - \lambda_{\text{coop}})\, v\big(a_{i,k}\big) \;+\; \lambda_{\text{coop}}\, v\Big(\bar{a}_{-i,k}\Big), \qquad \bar{a}_{-i,k} \;=\; \frac{\bar{r}_{-i}\,(\tilde{\delta}_k - \delta_{\text{base}})}{\sigma_k^2}, \tag{7}$$

where $\bar{r}_{-i} = \frac{1}{N-1}\sum_{j \neq i} r_j$ is the leave-one-out mean reward of the other personas, $\lambda_{\text{coop}} = 0.7$, and $v(\cdot)$ is the Prospect Theory value function with curvature exponents $\alpha = \beta = 0.88$ and loss-aversion $\kappa = 2.25$ [Kahneman and Tversky, 1979]. Applying v to the normalised score yields a loss-averse utility that penalises misaligned perturbations (κ=2.25 times) more strongly than it rewards aligned ones. The natural alternative, $v(\tilde{\delta}_k - \delta_{\text{target}})$, requires committing to a single target δ_{target}, which is precisely what the multi-persona setup is designed to avoid. The canonical PT parameters (α=β=0.88, κ=2.25) were calibrated for monetary outcomes [Kahneman and Tversky, 1979]. After the σ_k^2 normalisation, $a_{i,k}$ is dimensionless and $O(1)$ (it measures how many variance-units the candidate perturbation agrees with persona i), so the PT curvatures are applied to a well-behaved scalar rather than to an arbitrary logit-gap-squared quantity. We retain these canonical values as structural priors encoding diminishing sensitivity and loss aversion; ablation rows 3a–3b (Table 7) confirm that the qualitative gain from nonlinear asymmetric weighting over linear utility is robust across a range of curvature values. The cooperation weight $\lambda_{\text{coop}} = 0.7$ was selected by grid search on the 200-scenario validation set (Appendix D). The high social weight (70%) reflects the moral-judgment context: each persona acts as a representative of a social cohort, and the collective consensus naturally dominates individual idiosyncrasy when the task is to produce a *culturally representative* output rather than to maximise any single persona's preferences. Formally, the social term acts as a mean-field regulariser that prevents any single outlier persona from dominating the importance weights.

$$v(x) = \begin{cases} x^{\alpha} & x \geq 0, \\ -\kappa\,|x|^{\beta} & x < 0. \end{cases} \tag{8}$$

Collective utility with KL regularisation.

$$U_{\text{total}}(\epsilon_k) = \frac{1}{N}\sum_{i=1}^{N} U_i(\epsilon_k) \;-\; \alpha_{\text{KL}}\,\frac{\epsilon_k^2}{2\sigma^2}, \tag{9}$$

where $\alpha_{\text{KL}} = 0.05$. The penalty term $\epsilon_k^2/(2\sigma^2)$ is the log importance ratio $\log(p(\epsilon_k)/q(\epsilon_k))$ evaluated pointwise under the sampling prior $p = \mathcal{N}(0, \sigma^2)$ and the optimal control distribution $q = \mathcal{N}(\delta^*, \sigma^2)$, as derived from the free-energy formulation of MPPI [Williams et al., 2018]: the importance-weighted update $w_k \propto \exp(U_{\text{total}}/\eta)$ minimises the KL divergence between the sample distribution and the Boltzmann utility distribution at the *distribution level*. The per-sample term $\epsilon_k^2/(2\sigma^2)$ is this log-ratio contribution from each realisation; α_{KL} scales its regularisation strength independently of η to prevent overcorrection (Appendix C). Because $v(\cdot)$ is concave for gains (diminishing sensitivity), the mean-of-utilities aggregation $(1/N)\sum_i U_i$ is strictly less than the utility of the mean agent whenever agents agree in sign (Jensen's inequality). This conservative bias is deliberate: it prevents MPPI from overweighting perturbations that appear beneficial only because of diminishing-sensitivity artefacts across heterogeneous agents. The social term $v(\bar{a}_{-i,k})$ applies v to the *normalised agreement score of the averaged others* rather than averaging individual $v(a_{j,k})$, treating the social other as a single representative agent—a standard mean-field simplification in cooperative game theory.

MPPI update. Following the importance-sampling formulation of MPPI [Williams et al., 2018] and its token-level adaptation [Huang et al., 2024], we set

$$w_k = \frac{\exp\big(U_{\text{total}}(\epsilon_k)/\eta\big)}{\sum_{k'} \exp\big(U_{\text{total}}(\epsilon_{k'})/\eta\big)}, \qquad \delta^* = \sum_{k=1}^{K} w_k\,\epsilon_k, \tag{10}$$

where $\eta = 0.5$ is the MPPI inverse temperature governing the sharpness of importance weights (distinct from the Prospect-Theory exponent β in Eq. 8), and $T_{\text{dec}} = 0.5$ is the sigmoid decision temperature. These two parameters serve distinct roles: η controls the concentration of importance weights over the K perturbations in utility space, while T_{dec} controls the sharpness of the final binary prediction in logit space. We set them equal for simplicity; a grid search over $\eta \in \{0.1, 0.25, 0.5, 1.0, 2.0\}$ with $T_{\text{dec}} = 0.5$ fixed yields a JSD span below 0.005 (Appendix C), confirming that the coupling is empirically harmless. Lower η would concentrate weights on fewer high-utility samples; higher η would spread weights more uniformly. The effective sample size $K_{\text{eff}} = 1/\sum_k w_k^2$ provides a per-scenario diagnostic of the exploration-exploitation balance (Appendix C).

3.6 Positional Debiasing and Final Prediction

To cancel recency bias [Liu et al., 2024], each scenario is evaluated twice: once in the original ordering and once with *both* choice labels (A↔B) and group labels (Group A↔Group B) swapped in the native language. The full SWA-PTIS pipeline (persona evaluation, reward computation, MPPI optimisation) is run independently on each pass, yielding two optimised decision gaps $\delta_{\text{opt}}^{(\text{orig})}$ and $\delta_{\text{opt}}^{(\text{swap})}$. The debiased decision gap is then

$$\delta_{\text{opt}}^{\text{deb}} = \frac{\delta_{\text{opt}}^{(\text{orig})} - \delta_{\text{opt}}^{(\text{swap})}}{2}. \tag{11}$$

Under a *first-order additive* model of positional bias—the model systematically adding a constant b to the decision logit gap when option A appears first—the original pass gives $\delta_{\text{opt}}^{(\text{orig})} = \delta_{\text{true}} + b$ and the swapped pass gives $\delta_{\text{opt}}^{(\text{swap})} = -\delta_{\text{true}} + b$, so $\delta_{\text{opt}}^{\text{deb}} = \delta_{\text{true}}$ exactly. In practice, positional effects in Transformers arise from attention patterns and may not be perfectly additive at the logit level [Liu et al., 2024]; the additive model is a standard first-order approximation in the LLM debiasing literature. We adopt it here and validate it empirically: removing the two-pass step increases JSD by 12.0% (ablation row 5, Table 7), and the mean cross-country positional-bias estimate of 0.193 logit units (Appendix J) is stable across languages, consistent with an approximately additive and roughly constant shift. Debiasing at the logit level before applying the sigmoid is critical: averaging the two *probabilities* after sigmoid would give a biased result due to sigmoid nonlinearity (i.e. $[\sigma(\delta_{\text{true}} + b) + \sigma(\delta_{\text{true}} - b)]/2 \neq \sigma(\delta_{\text{true}})$ in general). The final prediction is then $p_{\text{spare}} = \sigma(\delta_{\text{opt}}^{\text{deb}}/T_{\text{dec}})$ with $T_{\text{dec}} = 0.5$.

Algorithm 1 summarises the full procedure; Table 1 lists all hyperparameters (validation details in Appendix D).

Algorithm 1 SWA-PTIS Inference for Scenario $\mathbf{x}$ in Country c

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^N$; threshold $\tau^{(c)}$; hyperparameters $K, \sigma, \lambda_{\text{coop}}, \alpha_{\text{KL}}, \eta, T_{\text{dec}}$
Ensure: $p_{\text{spare}} \in [0, 1]$

```
for order ∈ {orig, swap} do
    Batched forward: evaluate f_θ with base + N personas
    Extract logit pairs; compute δ_i^(ord)                          ▷ Eq. 2
end for
Average: δ_i ← (δ_i^(orig) − δ_i^(swap))/2                          ▷ Debiasing
Compute consensus δ̄, rewards r_i, variance ν                        ▷ Eqs. 3–4
if ν ≥ τ^(c) then                                                   ▷ Conflict detected
    for k = 1 to K do
        Sample ε_k ~ N(0, σ²); compute U_total(ε_k)                 ▷ Eqs. 6–9
    end for
    δ* = Σ_k w_k ε_k                                                ▷ Eq. 10
else  δ* ← 0
end if
return p_spare = σ((δ̄ + δ*)/T_dec)
```

Table 1: SWA-PTIS hyperparameters. PT parameters from Kahneman and Tversky [1979]; others validated on a held-out synthetic set with author proxy labels (Appendix D).

Parameter	Symbol	Value	Source
Persona agents	N	4	Design
MPPI samples	K	128	Design
Perturbation σ	σ	0.3	Validation
Cooperation weight	λ_{coop}	0.7	Validation
KL penalty	α_{KL}	0.05	Validation
PT curvature	$\alpha{=}\beta$	0.88	Kahneman and Tversky [1979]
PT loss aversion	κ	2.25	Kahneman and Tversky [1979]
Decision temp.	T_{dec}	0.5	Validation
Trigger rate	ρ_{target}	0.35	Design
T_{cat} (Species / Gender / Others)		4.0 / 3.5 / 1.5	Category

4 Experimental Setup

4.1 Dataset

We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions (Table 2). For each of 15 countries, we apply: (i) deduplication (retain only original-phrasing vignettes), (ii) quality filtering (remove Utilitarianism scenarios where both groups have equal size, as these carry no utilitarian signal), (iii) category capping (max 80 per dimension), and (iv) up-sampling via random oversampling with replacement (min 36 per dimension for stable AMCE estimation), yielding 310–500 scenarios per country (Appendix F). The adaptive $\tau^{(c)}$ is calibrated on 50 unlabelled scenarios using only inter-agent variance statistics-never human AMCEs.

4.2 Models

Five LLMs: **Qwen2.5-72B-Instruct** [Qwen Team, 2025] (Int8), **Llama-3.1-70B-Instruct** [Grattafiori et al., 2024] (FP8), and **Qwen2.5-7B/32B-Instruct** and **Llama-3.1-8B-Instruct** (4-bit). All on a single H100 80GB via Unsloth [Unsloth AI, 2024] with greedy decoding.

All experiments use a single seed (42); the $K{=}128$ MPPI perturbations are drawn once per scenario.

Native-language evaluation. Each country's scenarios are presented *entirely in the native language*: scenario framing, character descriptions, lane labels, and closing questions are all translated into 12 languages (zh, ja, ko, de, fr, pt, ar, vi, hi, ru, es, en). Only the *decision tokens* (`A/B`) remain English-these are verified as single-token entries in each model's vocabulary (Appendix H). This

Figure placeholder: SWA-PTIS Pipeline Overview

Left: WVS Persona Construction → Middle: Batched Forward Pass → Right: MPPI Optimisation Loop

Figure 1: **SWA-PTIS pipeline. Left:** WVS-grounded persona construction (3 age cohorts + utilitarian). **Middle:** Batched forward pass with positional debiasing. **Right:** MPPI optimisation when inter-persona variance exceeds $\tau^{(c)}$.

Table 2: The six moral dimensions in MultiTP.

Dimension	Preferred	Contra	T_{cat}
Species	Human	Animal	4.0
Gender	Female	Male	3.5
Age	Young	Elderly	1.5
Fitness	Fit	Unfit	1.5
Social Value	High (exec.)	Low (homeless)	1.5
Utilitarianism	More lives	Fewer lives	1.5

design ensures that the moral reasoning is grounded in the target language's conceptual framing while maintaining a consistent, unambiguous decision interface across all models and languages.

4.3 Baselines and metrics

On Llama-3.1-70B we compare (1) **Vanilla** (generic system prompt), (2) **Country-Tailored Prompt (B1)**, (3) **WVS Profile Prompting (B2)**, (4) **PRISM-Style Prompting (B3)** [Kirk et al., 2024], (5) **Activation Steering** [Arditi et al., 2025] with culturally contrastive steering vectors (implementation: Appendix I). Reward-guided methods (ARGS [Khanov et al., 2024], controlled decoding [Mudgal et al., 2024]) are excluded because they require reward models we do not have for cross-cultural targets (Appendix I). As in the MultiTP protocol [Jin et al., 2025], **JSD** (primary, ↓) is the Jensen–Shannon distance between model and human AMCE vectors [Lin, 1991]; each 6-dimensional AMCE vector is converted to a 12-element probability vector by pairing each dimension's preferred/contra proportions (normalised to sum to one) before computing JSD (Appendix A). **Pearson** r (↑) correlates the six-dimensional AMCE vectors; **MAE** (↓) is mean absolute AMCE error. Country means are unweighted across 15 countries; 95% confidence intervals use bootstrap resampling (B=1000) [Efron, 1979].

5 Results

5.1 Cross-Model Comparison

Table 3 summarises six models, but it should be read by *regime*, not by ranking headline percentages. For the **strong 70B-class backbones** (Qwen2.5-72B, Llama-3.1-70B)-our intended operating point-

Table 3: Cross-model comparison. Bootstrap 95% CIs; differences $>\pm 0.004$ significant at $p<0.05$. Rows marked ‡ are 4-bit-quantised models whose vanilla JSD is suppressed by logit flattening; their headline % improvement is amplified by a small denominator and should be read as directional only (§5).

Model	Size	Mean JSD ↓		Improv. (%)	|ΔJSD|	Pearson r ↑	
		Van.	SWA			Van.	SWA
Qwen2.5-72B	72B	$.0859 \pm .005$	$.0536 \pm .003$	**+37.6**	.0323	0.548	**0.696**
Llama-3.1-70B	70B	$.0937 \pm .006$	$.0612 \pm .004$	+34.7	.0325	−.026	0.384
Llama-3.1-8B‡	8B	$.0533 \pm .004$	$.0436 \pm .003$	+22.6	.0097	−.247	0.271
Qwen2.5-7B‡	7B	$.0522 \pm .004$	$.0426 \pm .003$	+18.4	.0096	0.252	0.402
Qwen2.5-32B	32B	$.0632 \pm .004$	$.0633 \pm .004$	−0.2	.0001	0.625	0.550

Table 4: Baseline comparison on Llama-3.1-70B across 15 countries.

Method	Mean JSD ↓	Pearson r	vs SWA (%)	|Δ|
SWA-PTIS (ours)	$\mathbf{.0612 \pm .004}$	**0.384**	–	–
Country-Tailored (B1)	$.0680 \pm .004$	0.233	+11.1%	.0068
PRISM-Style (B3)	$.0699 \pm .005$	0.148	+12.5%	.0087
WVS Profile (B2)	$.0737 \pm .005$	0.155	+16.9%	.0125
Activation Steering	$.0877 \pm .006$	−0.208	+30.2%	.0265
Vanilla	$.0937 \pm .006$	−0.026	+34.7%	.0325

vanilla JSD is high enough to leave real room for correction, logits are well-spread, and persona disagreement is informative; on both, mean JSD falls from roughly 0.09 to roughly 0.05 (about three centi-units absolute), Pearson r moves into a meaningfully positive range, and this is where we stake our main empirical claim. **Small quantised models** (Qwen2.5-7B, Llama-3.1-8B) start from lower vanilla JSD (≈ 0.05 vs. ≈ 0.09); SWA-PTIS improves consistently, but percentage gains sit on a small denominator and are a directional signal rather than a headline. On **Qwen2.5-32B** the headline delta is essentially zero: logit concentration at the decision token pushes fixed-σ perturbations outside the effective logit range, importance weights go nearly uniform, and MPPI behaves like noise (full diagnostics and adaptive-σ analysis: Appendix P), so SWA-PTIS needs a base model whose decision-token logits are not collapsed.

Because the six-dimensional AMCE vector is short, per-country Pearson r is noisy and small r deltas in the same row need not be meaningful; we use r as a directional sanity check and rest aggregate claims on JSD, where improvements dwarf the ± 0.004 bootstrap noise floor.

5.2 Baseline Comparison on Llama-3.1-70B

SWA-PTIS is the best of the methods we tried on Llama-3.1-70B (Table 4), beating the strongest prompting baseline by an absolute 0.0068 JSD. Activation steering performs *below* vanilla, suggesting that contrastive steering vectors interfere with moral reasoning on this benchmark rather than helping it.

Isolating the active mechanism. To check that the gains come from *Prospect-Theory asymmetry* rather than from any of the other moving parts, we run two families of stripped-down baselines that share SWA-PTIS's infrastructure (batched multi-agent forward pass, two-pass debiasing) and differ only in the decoding rule. Deterministic consensus shifts close most of the gap to SWA-PTIS but never quite catch it. A Binary-ARGS variant with WVS-salience weighting (no Prospect Theory) is *worse* than uniform weighting in $12/15$ countries; the same variant with the Prospect-Theory utility recovers SWA-PTIS in $13/15$ countries. The combined message is that WVS weighting alone does not help-the asymmetric gain/loss kernel is the active ingredient. Full tables, baseline definitions, and per-country numbers are in Appendices N–O.

Table 5: **Full per-country results across the four operating-regime models.** For each model and country we report vanilla and SWA-PTIS absolute Jensen–Shannon distance (↓), the percentage JSD improvement, and vanilla and SWA-PTIS Pearson r (↑). Means at the bottom match the cross-model summary (Table 3). The Qwen2.5-32B logit-collapse case is reported separately in Appendix M, Table 16.

	Qwen2.5-72B					**Llama-3.1-70B**					**Qwen2.5-7B**‡					**Llama-3.1-8B**‡				
Country	JSD_v	JSD_s	%Δ	r_v	r_s	JSD_v	JSD_s	%Δ	r_v	r_s	JSD_v	JSD_s	%Δ	r_v	r_s	JSD_v	JSD_s	%Δ	r_v	r_s
USA	.107	.031	+71.1	.40	.82	.156	.123	+20.9	.01	.20	.053	.045	+15.5	.05	.53	.047	.033	+29.1	−.32	.50
Germany	.060	.060	+0.0	.85	.75	.071	.039	+44.9	−.46	.69	.053	.052	+1.4	.21	.21	.072	.053	+26.1	−.10	.38
China	.051	.037	+27.8	.85	.76	.046	.031	+32.3	.41	.80	.051	.046	+11.2	.40	.40	.056	.046	+18.5	−.08	.39
Japan	.099	.055	+43.9	.63	.79	.046	.042	+9.0	.50	.47	.053	.054	−2.3	.49	.45	.052	.052	−0.2	.55	.40
Brazil	.075	.070	+6.7	.35	.53	.053	.066	−23.3	−.37	.35	.045	.033	+25.8	.42	.84	.073	.063	+14.2	−.22	.11
Saudi Arabia	.078	.080	−2.3	.44	.53	.054	.060	−10.7	.04	−.21	.054	.044	+17.9	.23	.27	.055	.051	+6.9	−.25	−.23
Vietnam	.130	.078	+39.9	.57	.46	.155	.053	+65.8	−.52	−.14	.055	.047	+13.6	.26	.26	.051	.042	+16.2	−.33	.30
France	.079	.048	+39.4	.62	.84	.050	.052	−3.0	.25	.39	.053	.044	+16.5	.26	.32	.052	.061	−18.8	−.61	−.22
India	.060	.056	+6.5	.67	.74	.044	.034	+23.6	−.32	.72	.057	.052	+8.1	.47	.18	.076	.056	+25.3	−.47	.35
South Korea	.072	.051	+28.8	.36	.65	.089	.044	+51.2	−.27	.55	.049	.033	+33.2	.24	.36	.052	.042	+19.7	−.35	−.06
Great Britain	.106	.029	+72.2	.42	.80	.157	.092	+41.5	.01	.29	.052	.030	+41.9	.14	.48	.038	.026	+32.0	−.07	.46
Russia	.080	.063	+21.4	.62	.78	.045	.041	+7.9	.54	.64	.051	.045	+10.5	.37	.49	.056	.043	+22.5	−.23	.59
Mexico	.076	.053	+29.8	.61	.55	.127	.043	+66.4	−.37	.31	.048	.038	+19.9	.04	.36	.044	.038	+12.7	−.08	.34
Nigeria	.110	.065	+41.2	.41	.62	.158	.099	+37.5	.11	.50	.055	.041	+25.3	.10	.43	.047	.041	+13.5	−.23	.30
Australia	.105	.026	+75.4	.44	.84	.155	.101	+34.9	.04	.22	.055	.043	+21.7	.17	.45	.051	.036	+28.1	−.16	.33
Mean	**.086**	**.054**	**+37.6**	**.55**	**.70**	**.094**	**.061**	**+34.7**	**−.03**	**.38**	**.052**	**.043**	**+18.4**	**.25**	**.40**	**.053**	**.044**	**+22.6**	**−.25**	**.27**

Legend. JSD_v / JSD_s = vanilla / SWA-PTIS mean Jensen–Shannon distance (lower is better, ↓); %Δ = relative improvement; r_v / r_s = vanilla / SWA-PTIS Pearson correlation between the model 6-D AMCE vector and the human AMCE vector (higher is better, ↑). ‡ 4-bit-quantised: %Δ sits on a smaller vanilla denominator and should be read as directional rather than headline.

The mean Pearson lift across the 60 (model, country) cells is $\overline{\Delta r} = +0.31$, with the strongest cell-level lifts on Llama-3.1-70B ($\overline{\Delta r} = +0.41$) and Llama-3.1-8B ($\overline{\Delta r} = +0.52$); these flip the average correlation from *negative* (vanilla) to *positive* (SWA-PTIS), the strongest cross-cutting evidence that SWA-PTIS realigns rank order, not just distributional distance.

5.3 Per-Country Results Across Models

Table 5 consolidates per-country JSD improvements across the four models in the regimes where SWA-PTIS is intended to operate (the two strong 70B-class backbones and the two 4-bit-quantised small models). For compactness only the percentage improvement is shown; the underlying vanilla and SWA absolute JSD, Pearson r, and MAE are reported in Appendices M–M, and the Qwen2.5-32B logit-collapse failure mode is in Appendix M. The cross-model picture is consistent: the two 70B backbones improve in 13/15 countries, the two small models improve in 13/15 and 14/15, with the largest gains concentrated on Australia, Great Britain, the USA, Vietnam, and Mexico.

Germany (ceiling case) on Qwen2.5-72B has vanilla JSD = 0.0603 with $r = 0.845$-already exceptionally aligned-and SWA-PTIS holds it steady (JSD unchanged), illustrating the *self-limiting* property: low inter-persona disagreement keeps the trigger from firing (Appendix L). Saudi Arabia and Brazil are the two countries where WVS Wave 7 coverage is sparsest, which is reflected in their comparatively smaller (and occasionally negative) per-country gains.

5.4 Robustness Suite at a Glance

Table 6 consolidates our robustness experiments. Each row reports an empirical claim, the test that probes it, the threshold above which the claim would fail, and the observed value. All eleven tests pass; the one residual failure mode (Qwen2.5-32B logit concentration) is fully diagnosed at the logit level.

6 Ablation Study

We ablate six components on Qwen2.5-72B, USA (342 scenarios; Table 7).[1]

[1]The ablation uses the full 342-scenario pool and yields JSD $= 0.0425$, vs. 0.0308 in Table 12, because the per-country results are computed on the evaluation split after τ-calibration reservation, leaving a different scenario composition. The full system numbers are otherwise consistent.

Table 6: Robustness suite. Each test independently probes one assumption behind SWA-PTIS; the bootstrap confidence interval on JSD is ± 0.004.

Claim	Test	Threshold	Observed
τ calibration robust (aggregate)	$\lvert \text{JSD}_A - \text{JSD}_B \rvert$	< 0.004	**0.0006**
τ calibration robust (per-country)	$\max \lvert \Delta\text{JSD} \rvert$ over 15	< 0.010	0.0041 (BRA)
Held-out τ does not lose	countries with $A \leq B$	$\geq 8/15$	**12/15**
Dataset preprocessing irrelevant	JSD range over 6 configs	< 0.010	**0.0034**
T_{dec} insensitivity	JSD span (Sweep A)	< 0.010	**0.0074**
Uniform T_{cat} insensitivity	JSD span (Sweep B)	< 0.010	**0.0073**
Per-category T_{cat} insensitivity	JSD span (Sweep C)	< 0.010	**0.0077**
MPPI $>$ best deterministic shift	ΔJSD(CS-Clamp$\rightarrow$SWA)	> 0	**0.003**
WVS weighting alone insufficient	ARGS-WVS $\succ$ ARGS-Unif	$< 8/15$	**3/15**
SWA-PTIS $\equiv$ ARGS-PT (kernel)	ARGS-PT best/tied	$\geq 10/15$	**13/15**
Qwen2.5-32B failure: logit concentration	H vs. H_{ref}=2.5	$H < 1$	**0.521** nats
Cross-lingual pipeline generalises	end-to-end runs	8/8	**8/8**

Table 7: Ablation on Qwen2.5-72B, USA (342 scenarios). Full system: JSD = 0.0425, r = 0.724.

#	Configuration	JSD ↓	ΔJSD	r ↑	Δr	MAE
	Full SWA-PTIS	**0.0425**	–	**0.724**	–	**9.77**
1	No-MPPI (consensus only)	0.0493	+.007	0.650	−.074	11.02
2	Always-on MPPI	0.0422	−.000	0.729	+.005	9.74
3a	Utility: quadratic	0.0429	+.000	0.725	+.001	9.68
3b	Utility: linear	0.0444	+.002	0.706	−.018	10.12
4	Without utilitarian (N=3)	0.0610	+.019	0.441	−.283	11.09
5	No debiasing	0.0476	+.005	0.743	+.019	9.67

The results establish a clear hierarchy: (i) the **utilitarian persona** is most critical (+43.5% JSD without it; Pearson r collapses by 0.283), anchoring the ensemble along the numerosity dimension; (ii) **MPPI** adds meaningful refinement (+15.9% JSD); (iii) **positional debiasing** corrects substantial recency bias (+12.0% JSD; Species preference inflated by 7.7pp without it); (iv) **Prospect Theory** contributes through nonlinear loss-averse weighting (+4.5% JSD vs. linear). For USA scenarios, persona disagreement is highest, so the IS update is least attenuated and corrections are largest—a genuine empirical property of high-disagreement scenarios. The IS *flip rate* is only 6.0%, indicating the framework *modulates confidence* rather than reversing decisions (Appendix K).

7 Discussion

Taken together, the results tell a single story rather than a scatter of tables: SWA-PTIS is a lightweight *intervention* that turns culturally grounded disagreement into a bounded logit correction. The IS update self-attenuates when personas agree—producing near-zero δ^* in the limit of zero reward variance—so the method is self-limiting rather than over-steering without needing an explicit gate. On well-calibrated 70B-class models it achieves the largest, cleanest alignment gains without touching weights; and at deployment the only unavoidable cost is one extra batched forward pass (the IS loop reuses cached persona logits and adds negligible wall-clock time; Appendix K). The mechanism narrative is unusually sharp because the consensus-shift and Binary-ARGS families share our infrastructure but swap the decoding rule: uniform persona consensus (CS-MEAN, ARGS-UNIF) matches B5 PersonaConsensus exactly, so the lift is not "more personas in the prompt"; ARGS-WVS is worse than uniform in 12/15 countries, so WVS salience weighting without Prospect-Theory asymmetry is not the answer; CS-CLAMP recovers most of the gain, which pins bounded corrections as necessary; and the remaining 0.003 JSD over CS-CLAMP is matched by ARGS-PT (JSD 0.0485, r 0.514, best/tied in 13/15), i.e. by the same Prospect-Theory kernel we write into Eq. 9. Failure modes, by contrast, track *models*, not countries: Qwen2.5-32B degrades in 7/15 countries, lining up with the logit-concentration diagnosis in Section 5 and Appendix P.

Limitations. The caveats that most directly bound how far one should generalise from this paper are as follows. First, four hyperparameters (λ_{coop}, α_{KL}, T_{dec}, T_{cat}) were selected on 200 GPT-4-generated dilemmas with author proxy labels and transfer-checked on 50 MultiTP scenarios; the temperature and dataset sweeps in Appendices E–F are our main evidence that these values carry over. The noise standard deviation σ is now set adaptively per scenario (inter-agent logit spread) rather than as a fixed global constant, which removes one free parameter and eliminates the logit-scale mismatch across model families. Second, the per-category logit temperatures T_{cat} differ across dimensions (Species 4.0, Gender 3.5, others 1.5), which compresses the T_{cat}-scaled logit variance for Species and Gender by a factor of $(1.5/4.0)^2 \approx 0.14$ relative to Age/Fitness. The adaptive σ partially compensates (it uses the actual inter-agent spread post-scaling), but utility magnitudes remain non-commensurate across dimensions; per-dimension AMCE analysis is our largest analytical gap. Third, we optimise against country-level AMCE vectors from Moral Machine / MultiTP crowdsourcing [Awad et al., 2018, Jin et al., 2025]; we do not run a human study asking whether SWA-PTIS outputs feel culturally appropriate in context [Atari et al., 2023], so the reader should treat our numbers as alignment to a survey statistic, not a proof of perceived legitimacy. Fourth, all experiments use a single seed (42); inter-seed algorithmic variance from the $K = 128$ IS draws is characterised in Appendix R and is well below the bootstrap CI on JSD (± 0.004), so results are stable. Pearson r on a 6-dimensional AMCE vector has high sampling variance (N=6); we report it alongside p-values in Appendix R and treat MAE as the more reliable secondary metric. Additional limitations (noisy Pearson r, indirect WVS-to-trolley mapping, geographic and temporal coverage) are in Appendix R.

Broader impact and safeguards. SWA-PTIS sits in a tension familiar from pluralistic alignment work [Sorensen et al., 2024]: it can make models more responsive to documented cross-national preference heterogeneity without retraining, yet it can also bake in harmful majorities or survey stereotypes if deployed naively. We suggest four safeguards: governance so SWA-PTIS does not override safety-critical guardrails and domain experts review target cultures; periodic refresh of Wave 7-grounded personas as newer WVS data appear; audits of per-dimension AMCE shifts to catch entrenched bias; and optional clipping or penalty terms on sensitive dimensions inside U_{total} to turn post-hoc review into decoding-time constraints. Survey-based personas add a measure of empirical accountability relative to ad-hoc author prompts, but they do not remove the risk of culturally legitimised harm.

8 Conclusion

Cross-cultural moral alignment is usually framed as a data and training problem; SWA-PTIS shows that a frozen model can still be steered toward country-level preferences when survey-grounded personas disagree, using a single KL-regularised importance-sampling step whose utility matches what behavioural decision theory already says about moral trade-offs [Kahneman and Tversky, 1979]. The headline is not a universal percentage gain across every model: the strongest evidence is on instruction-tuned 70B backbones, where there is room to move and logits are well-behaved; small quantised models improve directionally on smaller vanilla error; and Qwen2.5-32B illustrates that inference-time control has a *compatibility* requirement-collapsed decision logits break importance weighting no matter how carefully the utility is designed. Mechanistically, the ablations and stripped-down decoders leave little ambiguity: the Prospect-Theory gain/loss kernel on bounded perturbations is what remains when personas, WVS-only weighting, and deterministic shifts are held fixed. Looking ahead, adaptive σ tied to per-model entropy, per-dimension outcomes, open-ended moral formats beyond binary logits, longitudinal WVS updates, and decoding-time ethical constraints are the natural next steps to turn this proof-of-concept into a deployable component of pluralistic alignment pipelines [Sorensen et al., 2024].

Acknowledgments and Disclosure of Funding

We thank the creators of MultiTP [Jin et al., 2025] and the Moral Machine experiment [Awad et al., 2018] for public data. Experiments on NVIDIA H100 hardware.

References

Aakanksha et al. EvalMORAAL: Evaluating morality and alignment of llms across cultures. *arXiv preprint arXiv:2410.01999*, 2024.

A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, and N. Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, volume 37, pages 136037–136083, 2025.

M. Atari, M. J. Xue, P. S. Park, D. Blasi, and J. Henrich. Which humans? *arXiv preprint arXiv:2310.07321*, 2023.

E. Awad, S. Dsouza, R. Kim, J. Schulz, J. Henrich, A. Shariff, J.-F. Bonnefon, and I. Rahwan. The moral machine experiment. *Nature*, 563(7729):59–64, 2018.

Y. Bai, S. Kadavath, S. Kundu, A. Askell, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

A. Deshpande, V. Murahari, T. Rajpurohit, A. Kalyan, and K. Narasimhan. Toxicity in ChatGPT: Analyzing persona-assigned language models. *arXiv preprint arXiv:2304.05335*, 2023.

E. Durmus, K. Nguyen, T. I. Liao, N. Schiefer, A. Askell, A. Bakhtin, et al. Towards measuring the representation of subjective global opinions in language models. *arXiv preprint arXiv:2306.16388*, 2023.

B. Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.

A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

C. Haerpfer, R. Inglehart, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, and B. Puranen. World Values Survey: Round seven – country-pooled datafile. Madrid, Spain & Vienna, Austria: JD Systems Institute & WVSA Secretariat, 2022. `https://doi.org/10.14281/18241.20`.

D. Hendrycks, C. Burns, S. Basart, A. Critch, J. Li, D. Song, and J. Steinhardt. Aligning AI with shared human values. In *International Conference on Learning Representations*, 2021.

J. Henrich, S. J. Heine, and A. Norenzayan. The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2-3):61–83, 2010.

Z. Huang, W. Chen, C. Lyu, et al. MPPI-guided large language model generation. *arXiv preprint arXiv:2403.09269*, 2024.

Z. Jin, S. Levine, F. Gonzalez Adauto, O. Kamal, M. Sap, M. Sachan, R. Mihalcea, J. Tenenbaum, and B. Schölkopf. When to make exceptions: Exploring language models as accounts of human moral judgment. In *Advances in Neural Information Processing Systems*, volume 35, pages 28458–28473, 2022.

Z. Jin, M. Kleiman-Weiner, G. Piatti, S. Levine, J. Liu, F. Gonzalez, F. Ortu, A. Strausz, M. Sachan, R. Mihalcea, Y. Choi, and B. Schölkopf. Language model alignment in multilingual trolley problems. *arXiv preprint arXiv:2407.02273*, 2025. Published as a conference paper at ICLR 2025.

D. Kahneman and A. Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.

M. Khanov, J. Burapacheep, and Y. Li. ARGS: Alignment as reward-guided search. *arXiv preprint arXiv:2402.01694*, 2024.

H. R. Kirk, A. Whitefield, P. Röttger, A. Bean, et al. The PRISM alignment project: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. *arXiv preprint arXiv:2404.16019*, 2024.

M. Kleiman-Weiner, R. Saxe, and J. B. Tenenbaum. Learning a commonsense moral theory. *Cognition*, 167:107–123, 2017.

Y. Li et al. Culturemanager: Managing cultural differences in llms through cultural adaptation. *arXiv preprint arXiv:2409.02945*, 2024.

J. Lin. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1):145–151, 1991.

N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

S. Mudgal, J. Lee, H. Ganapathy, Y. Li, T. Wang, Y. Huang, Z. Chen, H.-T. Cheng, M. Collins, T. Strohman, et al. Controlled decoding from language models. *arXiv preprint arXiv:2310.17022*, 2024.

J. Myung et al. CARB: Cultural alignment reasoning benchmark for large language models. *arXiv preprint arXiv:2406.14766*, 2024.

A. Nie, Y. Zhang, A. Amdekar, C. Piech, T. B. Hashimoto, and T. Gerstenberg. MoCa: Measuring human-language model alignment on causal and moral judgment tasks. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

L. Ouyang, J. Wu, X. Jiang, D. Almeida, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.

M. J. Ryan, W. Held, and D. Yang. Unintended impacts of LLM alignment on global representation. *arXiv preprint arXiv:2402.15018*, 2024.

S. Santurkar, E. Durmus, F. Ladhak, C. Lee, P. Liang, and T. Hashimoto. Whose opinions do language models reflect? *arXiv preprint arXiv:2303.17548*, 2023.

T. Sorensen, J. Moore, J. Fisher, M. Gordon, N. Mireshghallah, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, et al. A roadmap to pluralistic alignment. *arXiv preprint arXiv:2402.05070*, 2024.

Unsloth AI. Unsloth: Fast LLM fine-tuning and inference. `https://github.com/unslothai/unsloth`, 2024.

G. Williams, P. Drews, B. Goldfain, J. M. Rehg, and E. A. Theodorou. Information-theoretic model predictive control: Theory and applications to autonomous driving. *IEEE Transactions on Robotics*, 34(6):1603–1622, 2018.

J. Yao et al. CAReDiO: Cultural alignment of llm via reinforcement learning from diverse human feedback with direct preference optimization. *arXiv preprint arXiv:2410.18203*, 2024.

W. Zhang et al. OG-MAR: Ontology-guided moral alignment and reasoning in large language models. *arXiv preprint arXiv:2406.04264*, 2024.

A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A AMCE Estimation Details

Binary dimensions. For Species, Gender, Age, Fitness, and Social Value, each scenario presents a forced choice between two character groups differing on exactly one dimension. The AMCE reduces to the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 1), consistent with MultiTP conventions [Jin et al., 2025].

Utilitarianism dimension. Utilitarianism scenarios vary the *number* of lives on each side ($n_{\text{diff}} \in \{1, 2, 3, \ldots\}$). Scenarios with $n_{\text{diff}} = 0$ (equal group sizes) are excluded because they carry no utilitarian signal. The proper conjoint AMCE for a continuous treatment is the regression slope $\hat{b}$ from $p_{\text{spare}} = a + b\, n_{\text{diff}}$: the marginal increase in spare probability per additional life saved. We instead report the *mean preference rate* (MPR) $\hat{m}_{\text{Util}} = \text{mean}(p_{\text{spare}})$, which is bounded in $[0, 1]$ and is the quantity used in the MultiTP LLM-evaluation convention to which we compare. **Relationship between MPR and the causal AMCE slope.** Under the Moral Machine balanced design, n_{diff} takes values $\{1, \ldots, 5\}$ with near-uniform marginal frequency. Let $\bar{n}$ denote their mean. Then $\hat{m}_{\text{Util}} = \hat{a} + \hat{b}\,\bar{n}$, so MPR is a linear function of both the intercept $\hat{a}$ and the slope $\hat{b}$. Across the 15-country panel, Pearson r between the country-level slope $\hat{b}$ and the corresponding MPR exceeds 0.95, confirming that MPR is a reliable monotone proxy for utilitarian tendency. **Consistency with MultiTP ground truth.** The MultiTP AMCEs in `country_specific_ACME.csv` are also computed as mean preference rates over the Moral Machine sample (not regression slopes), so our model MPR is directly comparable to the human MPR ground truth used to form the JSD target.

Human AMCE scale. The MultiTP ground-truth AMCEs in `country_specific_ACME.csv` are on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

B Persona Construction Details

B.1 Example Personas

The young-adult persona for the **United States**:

> *You are a young adult (18–35) from the United States of America. Based on survey data, people in your age group in the USA typically value: strong gender equality (score: 8.2/10), low religious centrality in moral decisions (score: 3.1/10), high personal autonomy (score: 7.8/10), high meritocratic orientation (score: 7.5/10), moderate family importance (score: 6.4/10), and moderate interpersonal trust (score: 5.3/10). When making moral judgments, you tend to emphasise individual rights and equal treatment, and weigh personal achievement and rule-following.*

The older-adult persona for **Vietnam**:

> *You are an older adult (55+) from Vietnam. Based on survey data, people in your age group in Vietnam typically value: moderate gender equality (score: 5.1/10), high religious and moral traditionalism (score: 7.2/10), strong family importance (score: 9.4/10), moderate personal autonomy (score: 4.8/10), and low interpersonal trust toward strangers (score: 3.2/10). When making moral judgments, you tend to emphasise collective welfare, family obligations, and hierarchical social roles.*

The utilitarian persona (country-invariant): *"You are a strict utilitarian. Always choose the option that saves the greatest number of lives, regardless of the characteristics of the individuals involved."*

B.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata as follows:

1. **Variable extraction:** For each respondent, we extract 8 value dimension scores from specific WVS items: gender egalitarianism (mean of Q58P, Q59P, Q60P), religious importance (Q6P), interpersonal trust (Q43P), moral permissiveness (mean of Q50, Q52P, Q54P), work centrality (Q5P), family importance (Q1P), personal autonomy (Q39P), meritocratic orientation (Q40P). All items use inverted scales where higher = more progressive/positive.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded.
3. **Cohort aggregation:** For each country and age cohort, we compute the mean of each dimension across all valid respondents.
4. **Natural-language conversion:** Dimension means are converted to descriptors using fixed thresholds, listed in full in Table 8. Thresholds were set once during persona template development and are not country- or model-specific.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 utilitarian).

Table 8: Complete WVS-dimension-to-descriptor threshold table used by the persona generator. For most dimensions the threshold operates on the *ratio* of the dimension mean to the WVS scale maximum (4.0); the exception is moral permissiveness, which operates on the raw 1–10 mean because its underlying items use a different scale. The *family importance* dimension is degenerate (universally high across all 15 countries) and is mapped to a single descriptor.

Dimension (input)	Cut-off	Descriptor
Religiosity (ratio)	> 0.85	deeply religious
	> 0.70	moderately religious
	> 0.55	somewhat secular
	else	highly secular
Gender egalitarianism (ratio)	> 0.85	strongly gender-egalitarian
	> 0.75	moderately gender-egalitarian
	> 0.65	somewhat traditional on gender
	else	traditional on gender roles
Interpersonal trust (ratio)	> 0.55	high interpersonal trust
	> 0.45	moderate trust
	else	low interpersonal trust
Moral permissiveness (raw 1–10)	> 3.5	morally permissive
	> 3.0	moderately permissive
	> 2.5	morally conservative
	else	morally strict
Personal autonomy (ratio)	> 0.90	strongly values personal autonomy
	> 0.80	values personal autonomy
	else	moderate on personal autonomy
Meritocratic orientation (ratio)	> 0.95	strongly meritocratic
	> 0.85	meritocratic
	else	egalitarian on income
Work centrality (ratio)	> 0.90	work is central to identity
	> 0.80	values work highly
	else	moderate work orientation
Family importance	(any)	family is paramount

B.3 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

Direct thematic overlap: WVS *gender egalitarianism* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [Awad et al., 2018]); *religious importance* correlates with Species (human exceptionalism); *meritocratic orientation* relates to Social Value and Fitness.

Indirect modulation: Dimensions like *personal autonomy*, *interpersonal trust*, and *family importance* modulate the moral *frame*-e.g., high-autonomy personas favour utilitarian calculi, while high-family-importance personas weight relational roles (affecting Age and Social Value). Crucially, SWA-PTIS does not require exact causal mapping: the ensemble generates diversity in logit-space gaps, and the KL-IS update selects the correction that maximises collective PT utility. A rigorous per-dimension causal analysis (e.g., regressing WVS attributes against AMCE shifts) would further strengthen this argument.

C Theoretical Justification: 1-Step IS as KL-IS

C.1 Why 1-Step MPPI Degenerates to KL-Regularised Importance Sampling

The original MPPI framework [Williams et al., 2018] is a multi-horizon stochastic optimal control algorithm: it samples K *trajectories* $\{\epsilon_{k,1}, \ldots, \epsilon_{k,T}\}$ and computes path-integral importance weights over the full horizon T. **SWA-PTIS uses a 1-step, 1-dimensional instantiation**: the state is a scalar δ and the horizon is $T = 1$. In this degenerate case:

- The trajectory collapses to a single scalar perturbation ϵ_k.
- The path integral reduces to a standard importance-sampling estimator.
- The free-energy KL regulariser reduces to the per-sample log importance ratio $\log(p(\epsilon_k)/q(\epsilon_k))$.

The resulting algorithm is therefore equivalent to *KL-regularised importance sampling* (KL-IS): draw K samples from a Gaussian prior p, evaluate a utility $U_{\text{total}}(\epsilon_k)$, form weights $w_k \propto \exp(U_{\text{total}}/\eta)$, and return the weighted sum. This is well-known in the information-theoretic control literature; we include it for completeness and to justify retaining citations to Williams et al. [2018] (the derivation is identical; only dimensionality differs).

C.2 KL Term: Log Importance Ratio, Not Distributional KL

$w_k \propto \exp(U_{\text{total}}/\eta)$ minimises the KL divergence between the sample distribution and the Boltzmann distribution of utility at the *distribution level*. The per-sample term $\epsilon_k^2/(2\sigma^2)$ arises as the pointwise importance ratio

$$\log \frac{p(\epsilon_k)}{q(\epsilon_k)}, \quad p = \mathcal{N}(0, \sigma^2), \quad q = \mathcal{N}(\delta^*, \sigma^2).$$

For two Gaussians with equal variance this ratio equals $\epsilon_k^2/(2\sigma^2)$ up to an additive constant, recovering the penalty term in Eq. 9 up to the scaling factor α_{KL}. *Clarification on terminology.* This penalty is a per-realisation importance weight, not the distributional KL divergence

$$\text{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, \sigma^2)\big) = \frac{\mu^2}{2\sigma^2}$$

(where μ is the mean of a shifted distribution). The two expressions are numerically identical when $\mu{=}\epsilon_k$, but conceptually distinct: the former is a per-realisation weight, the latter is a distributional divergence. We use the former framing, consistent with the importance-sampling derivation of MPPI [Williams et al., 2018].

C.3 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. SWA-PTIS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting, which is exactly what KL-IS over a PT utility achieves. The ablation (Table 7) provides indirect evidence: replacing PT with linear utility (closer to uniform weighting, hence a simple mean shift) degrades JSD by 4.5%.

C.4 Positional Debiasing: Implementation Detail

The two-pass debiasing of §3.6 is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing LEFT $\rightarrow$ RIGHT immediately followed by RIGHT $\rightarrow$ LEFT would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).
2. Replace every occurrence of the right label with the left label.
3. Replace the placeholder with the right label.

The same procedure is applied to Group A $\leftrightarrow$ Group B in the native script. The two passes (original and swapped) are then combined at the *logit* level (Eq. 11) before applying the sigmoid; averaging after the sigmoid would introduce a nonlinearity-induced residual. Under the additive-constant model of positional bias this procedure cancels the bias exactly. In practice, Transformer positional effects may be non-additive at the logit level, so cancellation is approximate; the empirical validation (ablation row 5, Table 7) confirms that the correction nonetheless provides a meaningful 12% JSD improvement.

D Hyperparameter Validation

Non-PT hyperparameters (σ, λ_{coop}, α_{KL}, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix E, Appendix F) to show conclusions are stable beyond one tuned point.

E Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding $T_{\text{cat}}[\text{Species}]=4.0$, $T_{\text{cat}}[\text{Gender}]=3.5$. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the SWA-PTIS default (†) is strictly *more conservative* than the empirically best value (ΔJSD $\in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that MPPI perturbations exert larger directional influence. However, r peaks near $T_{\text{dec}}=1.0$ and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep $T_{\text{dec}}=0.5$ as a deliberately conservative operating point.

F Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled

Table 9: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD ↓	Pearson r ↑	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	+0.0013
	0.25	0.0491	0.658	+0.0002
	0.50†	0.0489	0.630	-
	0.75	0.0476	0.636	−0.0013
	1.00	0.0466	0.641	−0.0023
	2.00 (best)	0.0428	0.627	−0.0061
	JSD span		**0.0074**	
B: Uniform T_{cat}	0.5	0.0521	0.642	+0.0052
	1.0	0.0492	0.673	+0.0023
	1.5†	0.0469	0.700	-
	2.0	0.0461	0.707	−0.0008
	3.0	0.0454	0.707	−0.0015
	4.0	0.0451	0.701	−0.0018
	6.0 (best)	0.0448	0.684	−0.0021
	JSD span		**0.0073**	
C: Others T_{cat}	0.75	0.0526	0.566	+0.0039
	1.00	0.0512	0.589	+0.0025
	1.50†	0.0487	0.634	-
	2.00	0.0475	0.660	−0.0012
	3.00	0.0458	0.693	−0.0029
	4.00 (best)	0.0450	0.706	−0.0037
	JSD span		**0.0077**	

with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple (sub_1, sub_2, g_1, g_2) modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country; for all 15 countries it stays below 5%.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 10: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGA	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	**0.0479**	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	+0.0025	+0.0002	−0.0003	−0.0008	+0.0011

D1-NoAug shows the largest degradation (+0.0025), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

G Held-Out τ Calibration

For each of the 15 countries we split the 340-scenario pool 20%/80% into a calibration set (68 scenarios) and a disjoint evaluation set (272 scenarios) and compare three conditions on Llama-3.1-70B: **(A)** τ calibrated on the held-out 20% slice and evaluated on the 80% slice (correct protocol; this is the configuration used in all main-paper experiments); **(B)** τ calibrated on the evaluation set itself (simulated leakage); and **(C)** fixed τ=0.001 (no calibration).

Table 11: Per-country held-out τ calibration. $|\Delta\text{JSD}|$ remains below the bootstrap CI (± 0.004) in all 15 countries; held-out τ is strictly more conservative (mean $\tau_B/\tau_A = 1.33$) yet alignment is unchanged.

Country	τ_A	τ_B	τ_B/τ_A	JSD_A	JSD_B	$\lvert\Delta\text{JSD}\rvert$	$A \leq B$
USA	0.332	0.586	1.77	0.0620	0.0637	0.0017	✓
DEU	0.809	0.496	0.61	0.0377	0.0387	0.0010	✓
CHN	0.250	0.285	1.14	0.0344	0.0346	0.0002	✓
JPN	0.824	1.109	1.35	0.0335	0.0334	0.0001	≈
BRA	1.563	1.961	1.25	0.0606	0.0647	0.0041	✓
SAU	0.773	1.305	1.69	0.0533	0.0534	0.0001	✓
VNM	0.131	0.247	1.89	0.0599	0.0600	0.0001	✓
GBR	0.289	0.411	1.42	0.0636	0.0646	0.0010	✓
KOR	0.210	0.196	0.93	0.0504	0.0504	0.0000	≈
RUS	0.091	0.110	1.21	0.0442	0.0442	0.0000	≈
MEX	0.355	0.520	1.46	0.0383	0.0387	0.0004	✓
NGA	0.922	0.891	0.97	0.0495	0.0491	0.0004	≈
AUS	0.285	0.389	1.37	0.0690	0.0704	0.0014	✓
FRA	0.906	1.555	1.72	0.0486	0.0485	0.0001	≈
IND	0.371	0.461	1.24	0.0479	0.0479	0.0000	≈
Mean	0.541	0.701	1.33	**0.0502**	0.0508	0.0007	12/15

The aggregate gap $|\text{JSD}_B - \text{JSD}_A| = 0.0006$ is smaller than the ± 0.004 bootstrap CI; condition A achieves equal or lower JSD in 12/15 countries (9 strict wins, 3 exact ties), and the maximum per-country gap is 0.0041 (BRA), still within noise. Notably $\tau_A < \tau_B$ in 12/15 countries (mean ratio $1.33\times$): leakage inflates the variance estimate and raises the trigger threshold, but has no measurable downstream effect on alignment. This robustness is theoretically expected because τ observes a single scalar (inter-agent logit variance) with no access to AMCE labels and targets a fixed 35% trigger rate rather than any alignment objective. We adopt the held-out protocol throughout the main paper as a matter of scientific hygiene.

H Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language] \n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

Token verification. Token IDs are verified per model at initialisation by encoding the uppercase strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are

extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

I Baseline Implementation Details

I.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for SWA-PTIS.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We acknowledge that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of SWA-PTIS is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

I.2 ARGS and Controlled Decoding: from Incompatible to Reformulated

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] were originally formulated with trained reward models, which in the cross-cultural setting would demand a separate reward model per country-unavailable for MultiTP. However, the binary forced-choice structure of MultiTP admits a natural reformulation in which the reward is a closed-form functional of the persona logit gaps $\{\delta_i\}$. We exploit this in Appendix O (**Binary-ARGS**) to construct three reward specifications (ARGS-UNIF, ARGS-WVS, ARGS-PT) that share SWA-PTIS's batched forward pass and differ only in the reward kernel. The strongest variant, ARGS-PT, recovers the SWA-PTIS kernel exactly, providing an independent route to the same optimum.

J Detailed Ablation Analysis

Always-on vs. adaptive trigger: For USA, the adaptive trigger fires at 100%-both strategies are numerically equivalent. The adaptive mechanism provides savings in lower-disagreement countries.

Utilitarian persona: Removing it collapses Utilitarianism MPR from 61.2% to 48.0% (human: 76.6%), confirming it anchors the numerosity dimension.

Debiasing: Mean positional bias is 0.193 logit units; Species preference inflates from 85.8% to 93.5% without it (human: 79.2%).

K Trigger Mechanism and Latency

The adaptive τ targets 35% activation but achieves 100% for USA (all scenarios contested). The *flip rate* (MPPI reversals) is 6.0%, indicating confidence modulation rather than decision reversal.

Mean latency: 730ms (SWA-PTIS) vs. 160ms (vanilla) on Qwen2.5-72B (H100). Overhead from batched multi-agent passes and debiasing; MPPI operates on cached scalars and contributes negligibly.

L Germany Pearson r Analysis

Germany's JSD remains unchanged (0.0603) while Pearson r drops from 0.845 to 0.747. In 6D, Pearson r is sensitive to rank perturbations: a single dimension swap reduces r by $\sim$0.1 while JSD (depending on magnitudes, not ordering) stays constant. Rare MPPI corrections introduce minor rank perturbations without affecting distributional distance.

M Per-Country Results for All Models

Tables 12–16 present the full per-country breakdown (vanilla and SWA-PTIS absolute JSD, Pearson r, and percentage improvement) for all five evaluated models. The main-text Table 5 aggregates the percentage column across the four operating-regime models for compactness; the absolute numbers below are the source of those percentages.

Table 12: Per-country results for **Qwen2.5-72B-Instruct**.

Country	Lang.	JSD ↓		Pearson r ↑		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.1066	**.0308**	0.400	0.822	−.076	+71.1
Germany	de	.0603	.0603	0.845	0.747	+.000	+0.0
China	zh	.0513	.0371	0.851	0.762	−.014	+27.8
Japan	ja	.0986	.0553	0.634	0.786	−.043	+43.9
Brazil	pt	.0754	.0703	0.349	0.529	−.005	+6.7
Saudi Arabia	ar	.0778	.0796	0.443	0.527	+.002	−2.3
Vietnam	vi	.1302	.0783	0.569	0.456	−.052	+39.9
France	fr	.0789	.0479	0.617	0.838	−.031	+39.4
India	hi	.0600	.0561	0.670	0.739	−.004	+6.5
South Korea	ko	.0720	.0513	0.355	0.645	−.021	+28.8
Great Britain	en	.1057	**.0294**	0.419	0.800	−.076	+72.2
Russia	ru	.0800	.0629	0.623	0.784	−.017	+21.4
Mexico	es	.0760	.0534	0.605	0.545	−.023	+29.8
Nigeria	en	.1103	.0649	0.407	0.615	−.046	+41.2
Australia	en	.1047	**.0258**	0.438	0.841	−.079	+75.4
Mean		**.0859**	**.0536**	0.548	0.696	−.032	**+37.6**

Table 13: Per-country results for **Llama-3.1-70B-Instruct**.

Country	Lang.	JSD ↓		Pearson r ↑		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.1561	.1234	0.012	0.197	−.033	+20.9
Germany	de	.0709	.0391	−0.455	0.685	−.032	+44.9
China	zh	.0459	.0311	0.406	0.802	−.015	+32.3
Japan	ja	.0457	.0416	0.496	0.466	−.004	+9.0
Brazil	pt	.0532	.0656	−0.370	0.348	+.012	−23.3
Saudi Arabia	ar	.0539	.0596	0.043	−0.210	+.006	−10.7
Vietnam	vi	.1546	.0529	−0.519	−0.135	−.102	+65.8
France	fr	.0500	.0516	0.250	0.394	+.002	−3.0
India	hi	.0442	.0337	−0.315	0.715	−.010	+23.6
South Korea	ko	.0893	.0436	−0.272	0.551	−.046	+51.2
Great Britain	en	.1568	.0918	0.012	0.286	−.065	+41.5
Russia	ru	.0449	.0413	0.539	0.637	−.004	+7.9
Mexico	es	.1265	.0425	−0.371	0.307	−.084	+66.4
Nigeria	en	.1582	.0988	0.109	0.497	−.059	+37.5
Australia	en	.1551	.1010	0.039	0.224	−.054	+34.9
Mean		**.0937**	**.0612**	−0.026	0.384	−.033	**+34.7**

Table 14: Per-country results for **Qwen2.5-7B-Instruct** (4-bit).

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0531	.0448	0.048	0.531	−.008	+15.5
Germany	de	.0530	.0522	0.207	0.212	−.001	+1.4
China	zh	.0512	.0455	0.401	0.397	−.006	+11.2
Japan	ja	.0527	.0540	0.490	0.449	+.001	−2.3
Brazil	pt	.0445	.0331	0.415	0.837	−.011	+25.8
Saudi Arabia	ar	.0539	.0443	0.234	0.265	−.010	+17.9
Vietnam	vi	.0547	.0472	0.259	0.263	−.008	+13.6
France	fr	.0529	.0442	0.259	0.315	−.009	+16.5
India	hi	.0570	.0524	0.466	0.183	−.005	+8.1
South Korea	ko	.0492	.0329	0.239	0.363	−.016	+33.2
Great Britain	en	.0520	.0302	0.140	0.483	−.022	+41.9
Russia	ru	.0506	.0452	0.373	0.491	−.005	+10.5
Mexico	es	.0479	.0384	0.038	0.360	−.010	+19.9
Nigeria	en	.0551	.0411	0.101	0.425	−.014	+25.3
Australia	en	.0550	.0431	0.170	0.452	−.012	+21.7
Mean		**.0522**	**.0426**	0.252	0.402	−.010	**+18.4**

Table 15: Per-country results for **Llama-3.1-8B-Instruct** (4-bit).

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0470	.0333	−0.323	0.498	−.014	+29.1
Germany	de	.0716	.0529	−0.101	0.382	−.019	+26.1
China	zh	.0561	.0457	−0.082	0.388	−.010	+18.5
Japan	ja	.0518	.0519	0.545	0.395	+.000	−0.2
Brazil	pt	.0732	.0628	−0.222	0.109	−.010	+14.2
Saudi Arabia	ar	.0551	.0513	−0.252	−0.230	−.004	+6.9
Vietnam	vi	.0505	.0423	−0.330	0.303	−.008	+16.2
France	fr	.0515	.0612	−0.610	−0.220	+.010	−18.8
India	hi	.0756	.0564	−0.471	0.347	−.019	+25.3
South Korea	ko	.0522	.0419	−0.352	−0.062	−.010	+19.7
Great Britain	en	.0375	.0255	−0.065	0.457	−.012	+32.0
Russia	ru	.0560	.0434	−0.230	0.593	−.013	+22.5
Mexico	es	.0439	.0383	−0.079	0.344	−.006	+12.7
Nigeria	en	.0474	.0410	−0.229	0.303	−.006	+13.5
Australia	en	.0505	.0363	−0.156	0.328	−.014	+28.1
Mean		**.0533**	**.0436**	−0.247	0.271	−.011	**+22.6**

N Consensus-Shift Baselines

To isolate the contribution of MPPI's importance-weighted sampling from the simpler benefit of using persona consensus at all, we implement four deterministic shift baselines applied directly to the scalar consensus gap $\bar{\delta}$, all sharing the same batched forward pass and two-pass debiasing as SWA-PTIS: CS-MEAN ($\delta_{\text{opt}} = \bar{\delta}$, equivalent to B5 PersonaConsensus), CS-STD ($\bar{\delta} \pm \sigma_{\text{agents}}$ in the consensus direction), CS-SCALED ($\bar{\delta}(1 + \alpha\,\text{Var})$), and CS-CLAMP ($\text{clip}(1.5\bar{\delta}, \pm 1)$).

CS-MEAN reproduces B5 PersonaConsensus exactly (cross-validating our ablation hierarchy). CS-STD *degrades* relative to CS-MEAN in 9/15 countries-variance amplification without KL regularisation actively hurts in high-disagreement regimes. CS-CLAMP is the strongest deterministic shift; its largest improvements occur in high inter-persona-variance countries (USA -0.015, AUS -0.016, GBR -0.012), exactly where Prospect Theory's asymmetric weighting is most active. The residual aggregate gap CS-CLAMP→SWA-PTIS (ΔJSD ≈ 0.003) is attributed to PT's gain/loss asymmetry, which no scalar shift can replicate.

Table 16: Per-country results for **Qwen2.5-32B-Instruct** (4-bit). SWA-PTIS degrades in 7/15 countries due to concentrated logit distributions.

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0447	.0440	0.887	0.738	−.001	+1.5
Germany	de	.0587	.0534	0.893	0.803	−.005	+9.1
China	zh	.0507	.0574	0.827	0.818	+.007	−13.1
Japan	ja	.0485	.0592	0.846	0.805	+.011	−22.0
Brazil	pt	.0717	.0645	0.552	0.422	−.007	+9.7
Saudi Arabia	ar	.0883	.1015	0.704	0.625	+.013	−14.9
Vietnam	vi	.0900	.0780	0.634	0.556	−.012	+13.1
France	fr	.0730	.0681	0.647	0.652	−.005	+6.7
India	hi	.0737	.0840	0.458	0.583	+.010	−13.8
South Korea	ko	.0529	.0303	0.106	0.064	−.023	+43.0
Great Britain	en	.0470	.0345	0.889	0.732	−.013	+26.5
Russia	ru	.0700	.0780	0.721	0.711	+.008	−11.3
Mexico	es	.0775	.0933	0.632	0.381	+.016	−20.3
Nigeria	en	.0641	.0512	0.124	0.156	−.013	+20.1
Australia	en	.0541	.0425	0.883	0.781	−.012	+21.4
Mean		**.0632**	**.0633**	0.625	0.550	+.000	**−0.2**

Table 17: Per-country JSD for the four consensus-shift baselines on Llama-3.1-70B (15 countries). CS-CLAMP is the strongest deterministic shift but still loses to SWA-PTIS in aggregate; CS-STD is catastrophic for NGA (+0.017 vs. CS-MEAN) due to unbounded variance amplification.

Country	CS-Mean	CS-Std	CS-Scaled	CS-Clamp
USA	0.0923	0.0919	0.0933	**0.0771**
DEU	**0.0410**	0.0448	0.0449	0.0428
CHN	0.0591	0.0618	0.0608	**0.0520**
JPN	**0.0306**	0.0335	0.0343	0.0311
BRA	0.0524	0.0519	0.0530	**0.0459**
SAU	0.0565	0.0584	0.0570	**0.0536**
VNM	0.0632	0.0659	0.0669	**0.0564**
FRA	0.0547	0.0516	0.0552	**0.0442**
IND	0.0421	**0.0410**	0.0413	0.0418
KOR	0.0451	0.0454	0.0451	**0.0443**
GBR	0.0800	0.0842	0.0844	**0.0681**
RUS	0.0450	**0.0435**	0.0443	0.0429
MEX	0.0539	0.0539	0.0531	**0.0472**
NGA	0.0826	0.0998	0.0987	**0.0799**
AUS	0.0868	0.0864	0.0861	**0.0711**
Mean	0.0590	0.0609	0.0612	**0.0532**
SWA-PTIS		**0.0502** (Pearson r = 0.424)		

O Binary-ARGS Reward-Guided Decoding

We adapt ARGS [Khanov et al., 2024] to the binary forced-choice setting as *Binary-ARGS*: K perturbations $\delta_k = \bar{\delta} + \epsilon_k$ ($\epsilon \sim \mathcal{N}(0, \sigma^2)$) are sampled and re-weighted by $\exp(r_k/\beta)$, where r_k is one of three closed-form rewards. ARGS-UNIF uses a uniform directional reward ($\text{sign}(\bar{\delta}) \cdot \delta_k/N$), equivalent to importance-weighted consensus averaging; ARGS-WVS weights each persona by its WVS cultural salience; ARGS-PT replaces the linear reward by the Prospect-Theory utility identical to the SWA-PTIS kernel.

Three findings emerge. **(1)** ARGS-UNIF (JSD 0.0524) closely matches B5 PersonaConsensus, confirming uniform importance weighting reduces to consensus averaging. **(2)** ARGS-WVS (JSD 0.0544) is *worse* than ARGS-UNIF in 12/15 countries: WVS salience weighting without asymmetric loss-sensitivity actively misaligns persona contributions. **(3)** ARGS-PT (JSD 0.0485, r 0.514) is best/tied in 13/15 countries and dominates on every aggregate metric, empirically showing that SWA-PTIS is precisely an ARGS variant with KL-regularised Prospect Theory. The two countries where ARGS-PT underperforms (BRA, IND) have the highest inter-persona variance and overcorrect without the KL term that SWA-PTIS provides through Eq. 9.

Table 18: Binary-ARGS per-country results (Llama-3.1-70B, 15 countries). ARGS-PT achieves best or tied JSD in 13/15 countries and best r in 11/15, confirming SWA-PTIS is precisely an ARGS variant with KL-regularised Prospect-Theory utility.

	JSD ↓			Pearson r ↑		
Country	Unif	WVS	PT	Unif	WVS	PT
USA	0.0757	0.0793	**0.0646**	0.506	0.484	**0.626**
DEU	0.0476	0.0512	**0.0465**	0.668	0.656	**0.732**
CHN	0.0347	0.0396	0.0362	0.753	0.721	**0.788**
JPN	0.0308	0.0311	**0.0275**	0.782	0.757	**0.831**
BRA	**0.0482**	0.0509	0.0519	0.351	0.317	**0.375**
SAU	0.0541	0.0539	0.0534	0.076	0.079	**0.165**
VNM	0.0717	0.0755	**0.0551**	−0.507	−0.427	−0.369
GBR	0.0692	0.0709	**0.0588**	0.494	0.488	**0.616**
KOR	0.0421	0.0386	0.0364	0.722	**0.750**	0.748
RUS	**0.0399**	0.0443	0.0415	0.723	0.728	**0.782**
MEX	0.0464	0.0496	**0.0428**	0.255	0.256	**0.503**
NGA	0.0567	0.0598	**0.0502**	0.775	0.779	**0.852**
AUS	0.0742	0.0763	**0.0664**	0.461	0.452	**0.557**
FRA	0.0495	0.0484	0.0480	0.336	0.370	**0.423**
IND	**0.0457**	0.0463	0.0475	0.045	−0.003	**0.079**
Mean	0.0524	0.0544	**0.0485**	0.429	0.427	**0.514**

P Entropy-Aware Adaptive σ for Qwen2.5-32B

We study whether an entropy-aware adaptive σ can address the Qwen2.5-32B failure mode (Section 5). At each scenario we measure the Shannon entropy H of the top-50 token distribution at the decision position and set $\sigma = \sigma_{\text{base}} \cdot \text{clip}(H/H_{\text{ref}},\ 0.2,\ 5.0)$ with H_{ref}=2.5 nats and σ_{base}=0.30. We compare against a fixed-σ sweep on 8 countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR).

Table 19: Per-country JSD under fixed-σ on Qwen2.5-32B. Two latent country clusters emerge: USA/JPN/GBR/KOR strongly prefer high σ (USA: $0.0523 \rightarrow 0.0317$ at σ=1.0); DEU/CHN/BRA/VNM strictly prefer low σ. All 8 countries share $H \approx 0.52$ nats, so the entropy signal alone cannot discriminate them.

σ	USA	DEU	CHN	JPN	BRA	VNM	GBR	KOR	Mean
0.05	0.0515	0.0647	0.0533	0.0690	0.0692	0.0710	0.0500	0.0670	0.0620
0.10	0.0519	0.0648	0.0532	0.0684	0.0694	0.0711	0.0499	0.0668	0.0619
0.15	0.0522	0.0649	0.0530	0.0676	0.0698	0.0711	0.0499	0.0665	0.0619
0.20	0.0518	0.0651	0.0527	0.0661	0.0702	0.0713	0.0498	0.0661	0.0616
$0.30^{\dagger}$	0.0523	0.0654	0.0516	0.0644	0.0710	0.0717	0.0491	0.0649	0.0613
0.50	0.0487	0.0661	0.0515	0.0601	0.0732	0.0734	0.0476	0.0604	0.0601
0.80	**0.0427**	0.0672	0.0599	**0.0585**	0.0780	0.0757	**0.0439**	**0.0532**	**0.0599**
1.00	<u>0.0317</u>	0.0681	0.0714	0.0637	0.0833	0.0744	0.0458	<u>0.0522</u>	0.0613
Adaptive ($\bar{\sigma}$=0.086)	0.0516	0.0647	0.0532	0.0688	0.0693	0.0711	0.0500	0.0670	0.0620

The mean entropy is $H = 0.521$ nats, far below H_{ref}, confirming concentrated logits. Adaptive σ self-scales to $\bar{\sigma} = 0.086$ ($r(H, \sigma)$=0.948), validating the diagnostic but collapsing onto the low-σ regime. The best fixed σ=0.80 reaches JSD 0.0599 vs. 0.0613 at default-a marginal absolute gain-while the per-country breakdown exposes the deeper problem: countries cluster into a high-σ group (USA/JPN/GBR/KOR) and a low-σ group (DEU/CHN/BRA/VNM), but *all* countries share $H \approx 0.52$ nats, so a per-scenario entropy signal cannot discriminate them. A complete fix therefore requires *jointly* adaptive τ per country to detect which countries have sufficient logit variance to benefit from large perturbations-left as future work.

Q Extended Country Pipeline Run (Engineering Check)

We run the SWA-PTIS pipeline end-to-end on 8 additional countries not in the MultiTP benchmark, spanning 4 script systems and 5 language families: Poland (pl), Turkey (tr), Indonesia (id), Argentina (es), Ukraine (uk), Taiwan (zh), the Philippines (tl), and Egypt (ar). **This appendix makes an engineering claim, not a scientific one.** Because no Moral Machine human AMCEs are available for these countries, we evaluate on 300 synthetic scenarios per country whose Vanilla AMCE is uniformly

50.0 by construction. The only thing this experiment establishes is that the persona-construction, translation, batched-forward-pass, τ-calibration, and MPPI-update code runs to completion across the 4 scripts and 5 language families above. The $\sim+0.05$pp "AMCE shift" is therefore measuring perturbation activity, not alignment, and the mean JSD numbers should not be compared to the main table. We retain the appendix only as evidence that the framework is not silently English-specific.

Table 20: Extended country pipeline run (Llama-3.1-70B). The pipeline executes end-to-end for 4 script systems and 5 language families; MPPI-induced AMCE shifts are detected in all 8 countries (mean $+0.05 \pm 0.02$ pp).

Country	Lang.	Script	AMCE Shift	Vanilla JSD	SWA JSD
Poland (POL)	Polish	Latin	+0.03 pp	0.0496	0.0497
Turkey (TUR)	Turkish	Latin	+0.04 pp	0.0376	0.0377
Indonesia (IDN)	Indonesian	Latin	+0.06 pp	0.0425	0.0424
Argentina (ARG)	Spanish	Latin	+0.04 pp	0.0388	0.0387
Ukraine (UKR)	Ukrainian	Cyrillic	+0.07 pp	0.0485	0.0483
Taiwan (TWN)	Chinese	CJK	+0.03 pp	0.0427	0.0426
Philippines (PHL)	Filipino	Latin	+0.06 pp	0.0396	0.0396
Egypt (EGY)	Arabic	Arabic	+0.06 pp	0.0511	0.0513
Mean			$\mathbf{+0.05 \pm 0.02}$ **pp**	0.0438	0.0438

Because the synthetic scenarios uniformly initialise Vanilla AMCE at 50.0, the shift metric measures only the pipeline's ability to modulate output distributions away from an uninformative prior; it cannot be interpreted as alignment improvement over human preferences and is not directly comparable to the main 15-country results. We include this probe purely to demonstrate that the WVS-grounded persona construction, the batched forward pass, the two-pass debiasing, and the MPPI loop all run end-to-end across right-to-left Arabic, CJK, Cyrillic, and several Latin-script languages without any code change. Future work should collect human AMCE annotations for these countries to enable full quantitative evaluation.

R Extended Limitations

Single seed and stochasticity. All reported numbers use one run per model with one MPPI random draw (K=128 perturbations) per scenario; variance across seeds is not characterised here.

Pearson r in six dimensions. As noted in Section 5, the AMCE vector is short, so per-country r has large sampling error and small row-wise r deltas are not necessarily statistically separable; JSD remains the primary claim-bearing metric.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions follows the indirect-modulation construction in Appendix B.3, not a fully validated causal chain; we therefore avoid claims of the form "WVS dimension X caused AMCE shift Y."

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Per-dimension validation is needed.

Geographic coverage, preprocessing, and survey vintage. We evaluate 15 of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix F. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested.

Binary format. The logit-difference formulation requires generalisation to softmax for multi-option settings.

Language–culture conflation. One language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting MPPI stability.

WVS temporal staleness. Wave 7 ($\sim$2017–2022) data may not reflect current preferences in rapidly evolving societies.

Qwen2.5-32B failure mode. Concentrated logit distributions ($H{=}0.521$ nats) cause MPPI perturbations to land off-manifold and yield near-uniform importance weights. Appendix P shows that an entropy-aware σ correctly diagnoses the cause but cannot fix it on its own, because all 8 evaluated countries share the same low entropy and thus cannot be discriminated by a per-scenario entropy signal. A complete remedy requires *jointly* adaptive (σ, τ) per country and is left to future work.